

FINAL PROJECT REPORT [WEEK 1]

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Group Number: Group 3

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Calendar Week: CW40

Date: [Oct 6–12, 2025]

Progress Summary

This week, our team received the components for both the **Auto** and **Manual** robots. We began by checking the usability of all electronic components due to the fact that many were found to be broken or damaged. After identifying these issues, we replaced the damaged parts with the working ones.

Next, we made a complete list of all metal parts received, because each part must be returned after the competition. Therefore, we started creating **CAD files** for every metal part we used in this project including some motor and servo used in this Auto robot.

For the robot design process, we prioritized the **Auto Robot** over the Manual one. We brainstormed design ideas and built initial prototypes, but we encountered limitations related to the rulebook specification, such as limitation of the robot's size and the number of motors and servos allowed.

By the end of this week, we came up with the design of the **drive frame** and the **cube collecting mechanism** (gate mechanism), and created an **assembly CAD file** for both the drive frame and gate mechanism.

Next Week Plan

- Begin detailed CAD for the full Auto Robot.
- Start testing motor placement and wiring layout.
- Begin early concept design for the Manual Robot.

Problems Encountered

- Several electronic components were unusable and had to be replaced.
- Design constraints due to limited size and motor count.
- Additional components are required for future design, including wheels, sensors, motors and additional screws.